

Partial Trace vs Tensor Product in Quantum Mechanics

1. Introduction

In quantum mechanics, composite systems are described using the **tensor product** of the Hilbert spaces of individual subsystems. If we have two subsystems A and B with Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, the joint system lives in:

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$$

We now explore the difference between **tensor product** and **partial trace**.

2. Tensor Product

The **tensor product** is the operation that *combines* two quantum systems into a joint system.

- If ρ_A is the density matrix of system A and ρ_B of system B , the joint density matrix for independent systems is:

$$\rho_{AB} = \rho_A \otimes \rho_B$$

- For pure states, if $|\psi_A\rangle \in \mathcal{H}_A$ and $|\phi_B\rangle \in \mathcal{H}_B$, the combined state is:

$$|\Psi_{AB}\rangle = |\psi_A\rangle \otimes |\phi_B\rangle$$

Key insight: Tensor product builds the full combined system. If A is a qubit and B is a qutrit, the tensor product makes a system of dimension $2 \times 3 = 6$.

3. Partial Trace

The **partial trace** is used to *reduce a composite system to a subsystem*, effectively ignoring one part.

- If the joint system is ρ_{AB} , the reduced density matrix of subsystem A is:

$$\rho_A = \text{Tr}_B(\rho_{AB})$$

where Tr_B denotes tracing over the Hilbert space of B .

- Mathematically, if $\{|b_i\rangle\}$ is a basis of $\mathcal{H}_B$:

$$\rho_A = \sum_i \langle b_i | \rho_{AB} | b_i \rangle$$

- Example for a 2-qubit system ($|0\rangle, |1\rangle$ basis):

$$\rho_{AB} = \frac{1}{2} (|00\rangle\langle 00| + |11\rangle\langle 11|)$$

Partial trace over B gives:

$$\rho_A = \text{Tr}_B(\rho_{AB}) = \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|)$$

Key insight: Partial trace is used to study subsystems, especially in entangled or mixed states.

4. Summary Table

Operation	What it does	Input	Output
Tensor product	Combines systems	ρ_A, ρ_B	$\rho_{AB} = \rho_A \otimes \rho_B$
Partial trace	Reduces to subsystem	ρ_{AB}	$\rho_A = \text{Tr}_B(\rho_{AB})$

Conceptual summary:

- Tensor product $\rightarrow$ builds up the system.
- Partial trace $\rightarrow$ reduces the system.

5. Multiple Choice Questions

Q1: Which operation creates a joint quantum system from two subsystems?

- A) Partial trace B) Tensor product C) Measurement D) Unitary transformation

Answer: B) Tensor product

Q2: What is the result of taking the partial trace over subsystem B of ρ_{AB} ?

- A) The full density matrix of the system B) The reduced density matrix of subsystem A
 C) A pure state of B D) The tensor product of ρ_A and ρ_B

Answer: B) The reduced density matrix of subsystem A

Q3: If $\rho_{AB} = \rho_A \otimes \rho_B$, what is $\text{Tr}_B(\rho_{AB})$?

- A) ρ_A B) ρ_B C) $\rho_A \otimes \rho_B$ D) Zero

Answer: A) ρ_A

Q4: Partial trace is especially useful for which type of states?

- A) Product states only B) Entangled or mixed states C) Classical states D) Coherent states only

Answer: B) Entangled or mixed states